ENGINEERING HANDOFF

Fixes and verification

StockPilot AI - web app, backend, database and imports

The implementation addresses all 36 non-mobile findings from the 41-item audit. Mobile findings MB-01 through MB-05 remain excluded by your instruction. Changes are local and uncommitted.

What changed

The work concentrates on reliable demo transactions, tenant isolation, correct stock and monetary calculations, recoverable Tally/Excel imports, and clear web feedback. Safe refusals now prevent unit changes and merges that would damage historical meaning.

Verification completed

• Backend: 312 tests passed, 0 failed; 1 opt-in database-copy replay test was skipped because its separate snapshot fixture was not configured.

• The backend run includes 16 guarded PostgreSQL cases for migrations, stock/cost persistence, invoice rounding, retry safety, tenant isolation and workspace cleanup.

• Web: all 60 tests passed, including rendered workflow interaction tests. Lint, TypeScript checks and the production webpack build passed.

• Source review: clean whitespace checks, no mobile changes, no staged changes and no commits. HEAD remains c2e7496.

• Runtime note: backend targets Java 21. Local tests ran on installed JDK 25 with the test agent compatibility flag; Docker continues to use Java 21.

Deployment status

The hosted upload fix is a configuration change: the Render Blueprint now declares a persistent disk and paid Starter service. No deployment or paid resource was created. File retention must still be verified after restart and redeploy on the actual host.

Limits of this result

Tests used isolated fixtures and a disposable PostgreSQL database. No live Tally connection, client database, hosted environment or end-to-end browser walkthrough was exercised. The original audit remains a historical record; this document adds implementation and verification status. Passing checks do not prove the absence of all defects.

01 / Web experience

Frontend fixes

Audit ID	Implemented behavior
FE-01	Demo transactions update a persisted stock ledger, calculated invoices, balances and dealer orders; records stay in their demo workspace.
FE-02	Changed accounts clear cached workspace data. Requests and late responses are blocked when the session changes across tabs.
FE-03	A saved invoice clears the form and generates a new invoice number for the next transaction.
FE-04	Business writes refresh related stock, invoices, reports and outstanding-balance queries.
FE-05	Growing lists and searchable selectors page through all records. Receivable totals use the full customer ledger.
FE-06	Queries and actions show loading, error and success feedback, including import review failures.
FE-07	Create, edit and destructive controls follow the authenticated user permissions.
FE-08	Payments accept decimal amounts; money displays retain two decimal places and restricted values stay hidden.
FE-09	Dealer carts support quantity corrections and removal before submission.
FE-10	Default transaction dates use the local calendar day.
FE-11	Monetary displays use the configured workspace currency, including the customer portal.
FE-12	Sale availability reads stock for the selected product and warehouse.

Database and persistence

Audit ID	Implemented behavior
DB-01	Purchase identity includes supplier and April-March financial year. Sales identity includes financial year. Checks and PostgreSQL unique indexes agree.
DB-02	Merge rejects incompatible units, costs and unsupported dependencies; compatible merges preserve unit conversions and invalidate derived forecasts.
DB-03	Payment, reminder update and retry receipt commit atomically. Matching operation keys replay the saved result without another payment.
DB-04	Smart Import holds a database lock and commits status with business effects. Interrupted work rolls back; retries reclaim legacy stuck markers only under the lock.
DB-05	Reset and deletion include Smart Import tables, payments, adjustments, snapshots, mappings and raw rows in foreign-key dependency order.
DB-06	Product and invoice unit costs retain 12 decimal places. The forward migration preserves existing stored values.
DB-07	Invoice lines round to two decimal places before aggregation. Request scale and stored invoice amount limits are checked before persistence.
DB-08	Staging, dry run and commit validate final-table field lengths, numeric limits and aggregate amounts before final writes.
DB-09	Forecast publication is serialized per tenant using a database transaction lock.
DB-10	Changing a base unit is blocked after business dependencies exist; unused product edits remain available.

Identity and precision notes

Invoice numbers are normalized for case and surrounding spaces. Existing conflicting normalized invoice identities need review before applying the new unique indexes. Historical values already rounded by the old schema are not reconstructed.

Imports and backend

Audit ID	Implemented behavior
IM-01	Overlapping import files keep voucher boundaries and compare content, avoiding duplicated or silently merged invoices.
IM-02	Excel imports accept typed date cells and supported text date formats.
IM-03	CSV parsing supports quoted commas, escaped quotes and multiline cells.
IM-04	Party matching reuses GSTIN identities and rejects conflicting identity data.
IM-05	Classic import keeps original rows for remapping, unknown-field debugging and row-level errors.

Audit ID	Implemented behavior
BE-01	Demand uses invoice-only sales as well as movement history, with linked invoice/movement events counted once.
BE-02	Margins and cost comparisons normalize source quantities to base units.
BE-03	Backend responses redact unauthorized cost/profit fields, including assistant evidence, reports and audit details.
BE-04	Reordering includes current-day demand and assigns incoming/outgoing quantities once to the appropriate warehouse.
BE-05	Existing stockouts are urgent; missing demand history is explicitly labeled.
BE-06	Partial payments leave the reminder open with its remaining amount; full payment closes it.

Setup and demo preparation

Audit ID	Implemented behavior
OP-01	The local guide now uses an explicit demo Compose override and separate project. Normal backend startup keeps demo seeding disabled.
OP-02	Render config uses /var/data/uploads on a persistent disk. Deployment documentation includes existing-file migration and restart/redeploy retention checks.
OP-03	The shared web client enforces request deadlines, handles session expiry for JSON and CSV, and describes uncertain writes. Payment and stock-adjustment retries use stable operation keys.

Before the client demonstration

1. Start the backend with Java 21 or the supplied Docker setup. Use a dedicated demo workspace and keep the web in real-backend mode for Tally/Excel import demonstrations.

2. Apply Flyway migrations V24 and V25 through normal startup against the intended demo database after reviewing its backup and invoice-identity preflight.

3. Use one representative client-approved file to rehearse staging, validation, preview, commit and the resulting stock/balance reports. Invoice OCR remains explicitly unavailable until a real extraction provider is configured.

4. Rehearse the owner and restricted-role views. Verify the deployed upload-retention check if the demo will run on Render.

Review locations

Backend: new integrity, idempotency, history and import validation services; V24/V25 migrations; regression tests under backend/src/test. Web: shared API/business helpers and reusable invoice, picker, feedback and pagination components. Operations: docker-compose.demo.yml, render.yaml and deployment/local testing guides.

Scope and repository status

No mobile files changed. No commit, push, production migration, deployment or client-data reset was performed. The original audit PDF and this verification PDF are under output/pdf/.